

Bark beetles and wildfires: How does forest recovery change with repeated disturbances in mixed conifer forests?

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Abstract. Increased wildfire activity and recent bark beetle outbreaks in the western United States have increased the potential for interactions between disturbance types to influence forest characteristics. However, the effects of interactions between bark beetle outbreaks and subsequent wildfires on forest succession remain poorly understood. We collected data in dry mixed conifer forests across Idaho and western Montana to test whether vegetation responses differ between sites experiencing single and repeated disturbances. We compared tree seedling density and age, surface fuel loading, and stand structure characteristics in stands that experienced either high severity wildfire, large-scale tree mortality from bark beetles, or stands that experienced high bark beetle mortality followed by severe wildfire within 3–8 years of attack. Tree seedling density was 300–400% higher in gray bark beetle-affected stands than burned sites, but there was no evidence that a beetle and wildfire interaction affected seedling densities. The age distribution of Douglas-fir and grand fir seedlings in stands with repeated disturbances differed from those that only experienced wildfire, suggesting that seed availability varies between these stands. Though both bark beetle outbreaks and wildfires resulted in the death of numerous large trees and surface woody fuel loads 100–200% greater than control sites, the creation of large snags and higher fuel loads across the landscape may have ecological benefits. Compounding effects of bark beetle activity and wildfires were not observed in surface fuel loadings or stand densities. Overall, the effects of high severity wildfire drove post-disturbance fuel complexes and succession whereas the effects of *Dendroctonus pseudotsugae* and *Dendroctonus brevicomis* outbreaks before wildfires resulted in minimal post-wildfire differences. We conclude that although seedling age structure is responsive to bark beetle and fire interactions, in terms of fuel complexes and tree densities these disturbances are non-additive and compounding effects on forest trajectory of dry mixed conifer forests of the northern Rockies were not supported.

Key words: bark beetle and fire interactions; *Dendroctonus brevicomis*; *Dendroctonus pseudotsugae*; dry mixed conifer forests; northwestern USA; repeated disturbances; wildfire.

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INTRODUCTION

The western United States has experienced numerous large fires (Westerling et al. 2006,

Littell et al. 2009) and extensive tree mortality from bark beetle (Curculionidae: Scolytinae) populations in recent decades (Raffa et al. 2008, Bentz et al. 2010, Meddens et al. 2012). Under-

standing changes in vegetative structure and tree seedling responses following disturbances is critical for understanding temporal patterns of long-term vulnerability and forest resilience. The increased frequency of extreme fire events and large-scale bark beetle mortality during the past decade (Gedalof et al. 2005, Bentz et al. 2010), combined with increasing likelihood of such events under climate change, has created a need to understand how interactions between these disturbance types may alter ecosystems. Although most research in this area examines the effect of bark beetle mortality on subsequent wildfire burn severity (defined herein as relative tree mortality), post-disturbance succession is also of widespread interest to ecosystem managers (Simard et al. 2011, Hicke et al. 2012, Harvey et al. 2013, Carswell 2014).

There are several major concerns about the impact of bark beetle outbreaks on subsequent fire and forest structure. First, bark beetle outbreaks alter forest structure and the distribution and spatial arrangement of surface and canopy fuels (e.g., Hoffman et al. 2012, 2015, Schoennagel et al. 2012). However, many of these changes are dependent on time-since outbreak, relative tree mortality and rate of mortality, which can feed back to influence tree regeneration and forest structure (Donato et al. 2013). Initially following a successful attack, the “red phase” occurs as needles will turn from green to red and can last one to four years. The “gray phase” follows, lasting up to a decade and is characterized by many standing dead trees but with no needles remaining in the canopy; (e.g., Simard et al. 2011, Hoffman et al. 2012, Schoennagel et al. 2012, Donato et al. 2013). The “old” or “silver” phase occurs once standing dead trees begin to fall and break, and lasts up to 30–40 years post-outbreak (e.g., Schoennagel et al. 2012, Donato et al. 2013). Previous researchers suggested, through simulation modeling, that bark beetle activity followed by wildfires interact to influence long term forest stand dynamics (Simard et al. 2011, Hoffman et al. 2013, Hoffman et al. 2015), yet, there is little empirical data that quantified these changes.

Second, it is presently unknown how repeated disturbances influence woody fuel accumulation and stand structure (Harvey et al. 2013). Shifts in stand structure and fuels are important consid-

erations for assessing potential fuel hazards, and coarse dead downed woody material and standing dead trees also play an important ecological role. Most tree killing bark beetle species attack mature seed-producing trees expediting forest structural changes upon successful colonization (McCullough et al. 1998, Bjorklund and Lindgren 2009). Additionally, wildfires increase the prevalence of both large and small diameter standing dead trees, especially in areas of high burn severity (Everett et al. 1999, Passovoy and Fule 2006). Dead woody material from mature trees can improve wildlife habitat for cavity nesting birds (Bull et al. 1997, Hutto 2006), create canopy gaps for tree seedling establishment sites (Takahashi et al. 2000, Kennedy and Quinn 2001) and are sources of organic material that enhance forest productivity (Graham et al. 1994). The ecological benefits that arise from large tree mortality may be lost when forest ecosystems experience a high severity wildfire shortly after beetle mortality, degrading ecosystem functionality.

Our collective understanding about the impact of repeated disturbances on vegetation response, defined here as forest structure and seedling establishment, is limited. Much of the recent research has involved fire-on-fire behavior interactions (e.g., Peterson 2002, van Wagtendonk et al. 2012, Parks et al. 2013, Teske et al. 2013), whereas the literature on post-fire effects and the impacts of different types of repeated disturbances is lacking. A substantial body of research investigates the effects of “time-since-disturbance” following wildfire (e.g., Passovoy and Fule 2006, Roccaforte et al. 2012, Stevens-Rumann et al. 2013) and bark beetle outbreak (e.g., Simard et al. 2011, Hoffman et al. 2012, Jolly et al. 2012, Donato et al. 2013, Hoffman et al. 2015) independently, but little data exists about forest structure resulting from the combined effect of these two disturbance types. Most research on bark beetle-fire interactions to date addresses high elevation forests (e.g., Bebi et al. 2003, Klutsch et al. 2011, Kulakowski et al. 2013), while little is known about low elevation mixed conifer forests (Hicke et al. 2012). There is a need for forest type-specific research as high elevation spruce (*Picea*) and lodgepole (*Pinus contorta*) forests and low elevation ponderosa pine (*Pinus ponderosa*) and Douglas-fir (*Pseudotsuga menziesii*)

differ greatly in structure, fire regimes, bark beetle communities, and climate, making direct comparisons difficult. Though little is known about interacting disturbances in dry mixed conifer forest types, these forests are extensive across the western USA, comprising at least 50% of US Rocky mountain forests (Schoennagel et al. 2004, Hessburg et al. 2005, Baker 2009). Dry mixed conifer forests of the northern Rockies are dominated by ponderosa pine, Douglas-fir, and grand fir (*Abies grandis*) and are vulnerable to wildfires and bark beetle outbreaks following a century of fire suppression (Hessburg et al. 2005) and recent drought (Schoennagel et al. 2005). The only previous field study of interactions between disturbances from fire and bark beetles in dry conifer forests found that bark beetle outbreaks did not affect subsequent burn severity. However, low tree seedling densities on sites with repeated disturbances suggest legacy effects on forest succession (Harvey et al. 2013).

We investigated how repeated disturbances influence subsequent seedling response in dry-mixed conifer forests, examining four wildfires with adjacent and interior *Dendroctonus* bark beetle outbreaks to answer the following questions: (1) How do tree seedling density and age structure differ between sites experiencing a single disturbance (fire or bark beetles) compared to sites experiencing repeated disturbances (bark beetle and fire)? (2) How do stand structure and surface fuel loadings differ between undisturbed forests and those with high tree mortality from single or repeated disturbances? (3) How do bark beetle outbreaks followed by wildfire interact to influence surface fuels and seedling establishment?

METHODS

Site selection

Data were collected in dry mixed conifer forests in the interior northwestern US from the Continental Divide in Montana through Idaho. To determine potential plot locations, we used a combination of spatially explicit data sets describing wildfires, bark beetle outbreaks, and vegetation type. For wildfires, we analyzed two Monitoring Trends in Burn Severity (MTBS) products: the fire perimeters for each wildfire, and the pre-processed Relative differenced Nor-

malized Burn Ratio (RdNBR) spectral index data. To analyze the severity and extent of bark beetle outbreaks, we used USDA Forest Service Aerial Detection Survey (ADS) data. To determine vegetation type we used Environmental Site Potential (ESP), a LANDFIRE product based on the biophysical environment and standardized across the US, and selected only sites that were classified as Northern Rocky Mountain Dry-Mesic Montane Mixed Conifer Forests (Rollins 2009). The MTBS-derived fire perimeters are constructed by analysts who rapidly and manually digitize the Landsat imagery. Burn severity thresholds created for forested and non-forested ecosystems (Miller and Thode 2007), were manually identified for each fire using interpretations of dNBR and RdNBR, raw pre- and post-fire satellite imagery, plot data (if available), and the analysts' own judgment (Eidenshink et al. 2007). The ADS data layers are created by manually mapping from aircraft the tree mortality attributed to different bark beetles or other mortality agents. We used the ADS data to identify areas affected by Douglas-fir beetle (*Dendroctonus pseudotsugae*) and western pine beetle (*Dendroctonus brevicornis*). Sites lie within Climate Division 4 in Idaho and 1 in Montana (NOAA 2014). These climates are defined by cold winters (-6°C January mean) and warm summers (17°C July mean) with 58 cm mean annual precipitation (averaged over 1900–2008; NOAA 2014).

We used MTBS, ADS, and ESP layers to stratify potential plot locations in dry-mixed conifer forests within wildfires that occurred in 2007 in stands that experienced bark beetle outbreaks in the previous 3–8 years. The 2007 fire year was chosen because there were multiple large wildfires that met our a priori requirements of forest type and disturbance history that year. We selected burn severity thresholds from the RdNBR data to identify areas that met our burn severity criteria of approximately $>80\%$ tree mortality or "high severity". Additionally, we only focused on areas that experienced moderate to high bark beetle induced mortality (25–90% total tree mortality). Given the diverse tree species composition, consistently high tree mortality from bark beetle induced mortality was not attainable as bark beetles select for larger trees of a given species. For all samplings we used sites

for which burn severity, bark beetle caused tree mortality, from MTBS and ADS data, and ESP were verified through field observations. Of all visited sites, 92% were accurately determined by the three remote sensing data layers. On the remaining 8% of visited sites, ADS data had the highest level of inaccuracy, with several errors in the ESP for a given site as well. We did not collect data on the 8% of non-validated sites.

Based on information from local land managers we excluded sites that experienced commercial or non-commercial logging, tree planting, or other management treatments in the past 50 years, including post-fire rehabilitation and hazard tree removal. This was done to reduce the variability caused by the potential effects of management on both seedling density and fuel loading. The four wildfires that met our criteria were: the Rattlesnake (1292–2001 m) and Poe Cabin fires (1054–1711 m) on the Nez Perce–Clearwater National Forest, Rombo Mountain Fire (1419–2038 m) on the Bitterroot National Forest, and Middle Fork Complex (Lightning) Fire (1019–1399 m) on the Boise National Forest (Fig. 1). These fires burned in mixed severity and began in July of 2007 (Table 1).

Field methods

We sampled the effects of five disturbance types: (1) wildfire without prior bark beetle-induced tree mortality, (2) bark beetle-attacked stands in gray phase (3–8 years post-outbreak) when the wildfire occurred, (3) red phase (1–2 years prior to study), and (4) gray phase areas outside the fire perimeter, and (5) undisturbed (control) plots. Though bark beetle outbreaks at different phases are not different disturbances we treated these distinct time-since-disturbance periods as separate factors in the analysis. Within each treatment we randomly established eight to ten sites for a total of 40–50 sites per wildfire, or 180 plots across the four wildfires. Control sites were established in areas around the fire perimeter that had similar elevation, aspect and slope to disturbed plots. Pre-disturbance, all sites were dominated by mature ponderosa pine, Douglas-fir, and grand fir with a minor component on some sites of lodgepole pine, western larch (*Larix occidentalis*), and subalpine fir (*Abies lasiocarpa*). Slopes were highly variable on plots, ranging from 2–70% slope. Parent material on the

Poe Cabin Fire was loess mixed with basalt colluvium; both the Rattlesnake Fire and Rombo Mountain Fire had quartzite colluvium; and the Middle Fork Complex (lightning) Fire burned over quartz monzonite (Baker et al. 1983).

At each randomly located site we established a 0.04-ha circular plot and a 30-m long, variable-width belt transect. We recorded the slope, aspect, and elevation from the center of this circular plot. On the 30-m long transect, laid perpendicular to the elevation contour and following the dominant hillslope, total width of the belt transect varied between 1 and 10 m using similar methods to Droske (2012). Width was determined prior to sampling with the objective of sampling at least 30 tree seedlings of the most dominant species. If no or few seedlings were observed initially on the belt transect, a 10-m width was set as a default. Data were recorded for all tree seedlings within the belt transect (including species), distance between each primary branch node, and approximate age of each seedling determined by counting the total number of nodes. We also recorded distance to and species of the 10 nearest live seed sources from the center of the 30-m transect.

On each circular plot, we tallied all trees >5 cm DBH as alive or dead, including stumps, to estimate percent tree mortality induced by the fire. Pre-wildfire tree density was calculated by adding fire-killed stems together with live trees. We determined cause of tree mortality, either fire or bark beetle, prior to fire using similar methods to Harvey et al. (2013). Trees were recorded as ‘killed by bark beetles prior to fire’ if presence of exit holes on outer bark was detected, if fully excavated adult and larval galleries were present on the vascular cambium (>50% of bole circumference or remaining visible cambium), and if no needles were retained in the canopy which may indicate bark beetle induced mortality after wildfire. Dead trees with no evidence of pre-fire beetle activity or clear evidence of post-fire beetle activity were assumed to have been ‘killed by fire.’ We measured diameter at breast height (DBH) and recorded tree species of all living and dead trees (the latter referred to as snags) >5 cm DBH to quantify post-disturbance stand structure. We also quantified the density of standing dead trees >40 cm DBH because they provide greater ecosystem benefits and generally have

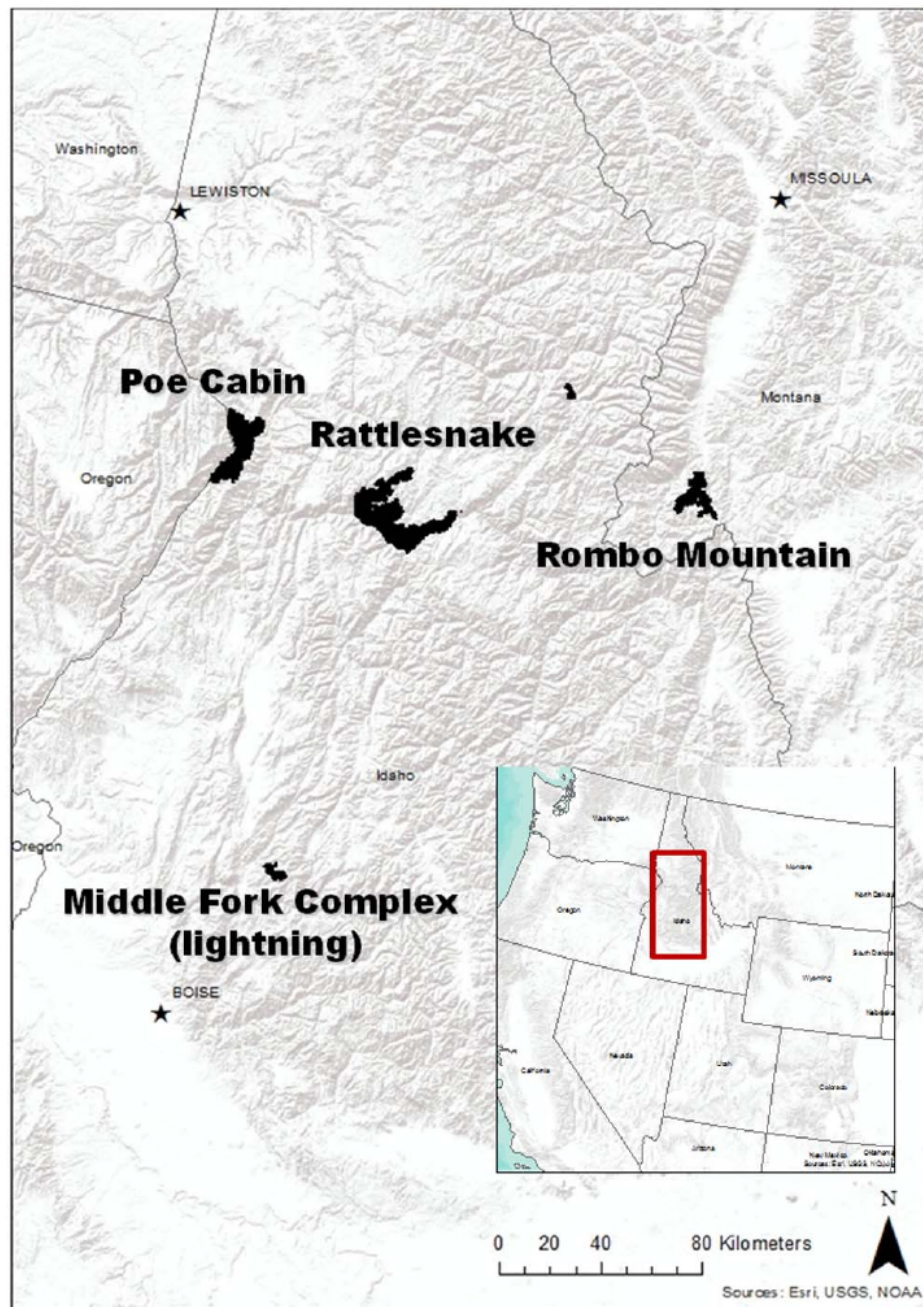


Fig. 1. Map of the four wildfires studied: Poe Cabin, Rattlesnake, Rombo Mountain, and Middle Fork Complex (Lightning) fires in central and northern Idaho and western Montana. All were in dry, low elevation mixed conifer forests in areas with prior bark beetle-induced tree mortality as mapped by the Aerial Detection System.

Table 1. Study fire information: fire dates, fire size, and percent in each burn severity classification.

Fire	Fire start date	Size (ha)	Unburned to low (%)	Low (%)	Mod (%)	High (%)	Area mask (%)†
Rombo Mountain	30 July 2007	11,418	26	15	13	16	30
Middle Fork (Lightning)	17 July 2007	2,851	33	22	16	6	23
Poe Cabin	18 July 2007	24,160	8	35	38	19	0
Rattlesnake	13 July 2007	41,485	36	35	16	13	0

† Non-processing area masks: areas in either pre-fire or post-fire reflectance imagery containing clouds, snow, shadows, smoke, significantly sized water bodies, missing lines of image data, etc.

the longest retention times within forested ecosystems (Bull 1983, Bull et al. 1997, Hutto 2006).

To quantify down woody debris, we measured fine woody debris and litter and duff depth on the 30-m transect using methods established by Brown (1974). Woody debris was broken up into size classes (0–0.64, 0.65–2.54, 2.55–7.62, >7.62 cm diameter), to correspond with 1-, 10-, 100-, and 1000-hr time-lag classes. The 1- and 10-hr fuels were measured along 6 m of transect and 100-hr fuels were tallied on a 10-m section of the transect. The 1000-hr fuels (coarse woody debris (CWD); logs ≥ 7.62 cm) were measured in 100-m² circular subplot located within the center of the overstory plot. Within this subplot the length and diameter at both ends was measured for every 1000-hr fuel and the fuel load was estimated following Keane et al. (2012). The 1000-hr fuels were also categorized by sound and rotten classes (Fosberg, 1970).

Data analysis

To test differences of fuel variables (pre-disturbance density, live tree density, large snag density, 1-, 10-, 100-, 1000-hr dead down and woody fuel loads, litter and duff depth) between disturbances we used a mixed effects analysis of variance (ANOVA), with site as a blocking variable ($df = 3$) and disturbance treatment as the main effect ($df = 4$). We checked assumptions of normality and equal variances through scatter plots of response variable residuals. To assess significant differences among disturbance classes when significance was observed in the overall test, Tukey's HSD test was used. Additional variables such as slope, aspect, and elevation were tested as covariates for all fuels categories, but were not found to be important. Finally, to test the influence of percent bark beetle mortality on fuel loading variables and seedling density we used percent bark beetle mortality as a continu-

ous variable within each of the three disturbance types that experienced bark beetle mortality (red, gray, and bark beetle followed by wildfire). Analysis was performed in JMP (SAS Institute 2007) with $\alpha = 0.05$.

To examine differences in the combined fuel complexes and tree seedling density among disturbance types we used a Multi-response Permutation Procedures (MRPP; Mielke and Berry 2001) in an Excel macro developed by the Rocky Mountain Research Station Statistician. Live tree density (>5 cm DBH), large snag density, biomass of 1-, 10-, 100-, 1000-hr fuels, litter depth, duff depth and log-transformed seedling densities as response variables and grouped by disturbance type, with fire as the blocking variable in the analysis were included. Pre-analysis, data were standardized by dividing each observation by the mean for that variable. When overall significant differences were detected we used Peritz Closure (Petrondas and Ruben 1983) multiple pairwise comparisons to separate disturbance type multivariate means.

We used total tree seedling density of all species to investigate differences in tree seedling establishment and age across disturbances, while age distribution analysis was performed on the three most abundant tree seedling species: ponderosa pine, Douglas-fir and grand fir. We log-transformed density data and added 1 prior to analysis to account for the non-normal distribution and zero-inflated nature of seedling density. Topographic variables such as slope, aspect, elevation and distance to seed source were analyzed as possible predictor variables in seedling density. Seedling age analysis was performed on the individual seedling rather than analyzing data at the plot level to accurately quantify the age structure of seedlings within a disturbance type rather than plot level averages. In these analyses we did not block by fire. The total number of tree seedling observations was

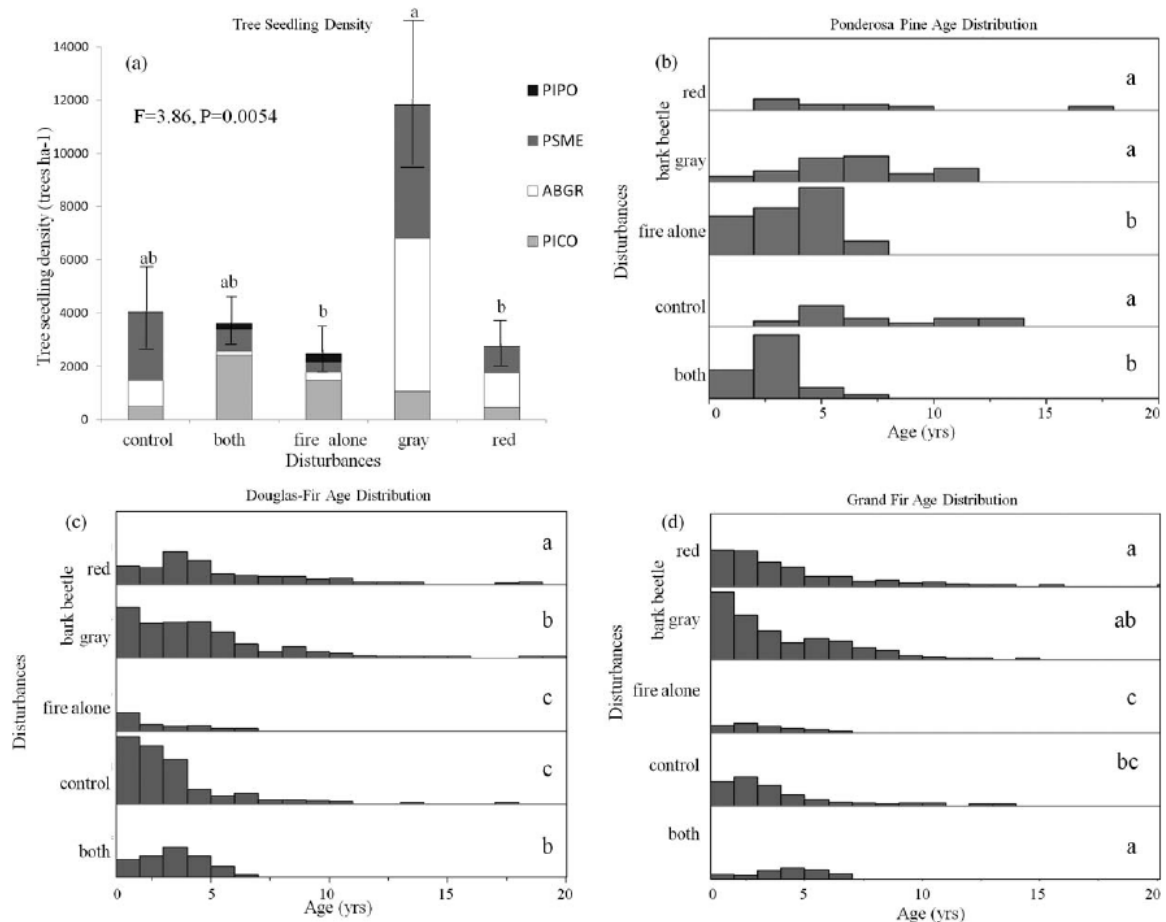


Fig. 2. Seedling density and age distribution by species. (a) Seedling density per hectare for each disturbance, broken up by the four most dominant species: lodgepole pine (PICO), grand fir (ABGR), Douglas-fir (PSME), and ponderosa pine (PIPO). Error bars indicate the standard error for each disturbance type and F and P values are the results from the ANOVA ($N = 180$). Histograms of seedling age distributions of (b) ponderosa pine ($N = 200$), (c) Douglas-fir ($N = 1275$), and (d) grand fir ($N = 1060$). Letters above or next to each bar indicate results from a Tukey's HSD test.

200, 1275, and 1060 for ponderosa pine, Douglas-fir, grand fir seedlings, respectively.

RESULTS

Tree regeneration

Tree seedling species were predominantly ponderosa pine, Douglas-fir, and grand fir, and one wildfire (Rattlesnake Fire) had a greater density of lodgepole pine. On less than 10 plots, Englemann spruce (*Picea engelmannii*) and subalpine fir seedlings were observed, comprising a minor component of total seedling density.

Seedling density varied significantly among disturbance types, with seedling densities four times greater in gray phase stands compared to all other disturbances (Fig. 2a). No statistically significant differences were observed between burned and control sites, although control sites had a slightly higher mean density of tree seedlings. We found no significant relationships between topographic variables such as aspect and slope and tree seedling density ($F = 1.79$, $P = 0.095$ and $F = 0.24$, $P = 0.63$, respectively). Seedling density was positively correlated with elevation and negatively correlated with distance

to seed source ($F = 8.62$, $P = 0.0039$; $F = 14.8$, $P < 0.0001$, respectively). However, these relationships were uniform across all disturbances and no interaction effect was observed. Seedling density was not significantly correlated with percent of bark beetle induced tree mortality in either the red ($F = 2.17$, $P = 0.15$, $R^2 = 0.07$) or gray red ($F = 0.45$, $P = 0.51$, $R^2 = 0.018$) phase stands or the bark beetle and fire affected stands red ($F = 1.81$, $P = 0.19$, $R^2 = 0.07$).

Tree seedling age structure differed not only by disturbance type but also by species. Ponderosa pine seedlings were a relatively minor component of the seedlings observed on control sites and bark beetle-affected sites, but densities were high on burned sites (Fig. 2a). Nevertheless, those present were significantly older than ponderosa pine seedlings on burned sites, with a mean age of six years on red phase bark beetle sites and up to seven years on control sites ($F = 20.73$, $P < 0.0001$; Fig. 2b). Ponderosa pine seedling ages were similar for bark beetle and fire-affected sites compared to fire-alone sites. We found more Douglas-fir and grand fir seedlings on control, red, and gray phase sites relative to ponderosa pine seedlings (Fig. 2c, d). Distributions of seedlings among disturbance types differed for both Douglas-fir ($F = 28.30$, $P < 0.0001$) and grand fir ($F = 4.16$, $P = 0.0024$). Though distributions on bark beetle affected sites were right skewed, both species had the oldest mean age on red phase sites, while gray phase sites had slightly younger mean seedling ages. Control and fire-alone sites had younger cohorts of seedlings for both species, but were not significantly different from one another. However, both Douglas-fir and grand fir also had a significantly older cohort of seedlings on bark beetle and fire-affected sites compared to fire-alone sites.

Fuel complex

Across all wildfires, pre-disturbance stand density did not vary by treatment type (Fig. 3a). Yet, post-disturbance live overstory tree density and basal area were significantly lower, if existent at all, on burned sites compared to unburned sites (Fig. 3b, c). Mean live tree densities of 265–281 trees ha^{-1} were observed in both red and gray phase bark beetle-affected stands compared to 450 trees ha^{-1} in control

stands. The densities of large snags were significantly higher in disturbed stands although no significant differences between bark beetle-affected stands (red and gray phase) and burned stands (Fig. 3d) were evident.

Overall, total downed woody debris was up to 200% greater on burned sites compared to unburned sites (Fig. 4). Although 10-hr and 100-hr fuels did not vary significantly among disturbances ($F = 1.90$, $P = 0.11$; $F = 1.33$, $P = 0.26$, respectively), there were slightly greater mean fuel loadings on burned sites. Conversely, fine woody debris (FWD; 1-hr fuels) and CWD were significantly different among disturbances (Fig. 4a, b). Rotten CWD was not a large component of the total CWD (<5%) and did not significantly vary among disturbances ($F = 1.70$, $P = 0.20$), thus sound and rotten CWD were pooled for all additional analysis. CWD and FWD fuel loadings were highest on bark beetle-affected areas in the gray phase, indicating the importance of time-since disturbance when compared to low fuels loadings in red phase stands. Similarly, high CWD loadings were observed in bark beetle and fire-affected stands, whereas fire-alone had slightly lower mean CWD loadings. Given the recent nature of the disturbance, red phase stands had CWD or FWD fuel loadings similar to control sites. Similar trends were observed in the analysis by percent bark beetle mortality. CWD and FWD loadings were positively correlated with percent bark beetle mortality in gray phase stands ($F = 5.54$, $P = 0.025$, $R^2 = 0.14$ and $F = 4.18$, $P = 0.049$, $R^2 = 0.11$), however no relationship was observed in red phase stands ($F = 2.14$, $P = 0.15$, $R^2 = 0.05$ and $F = 0.48$, $P = 0.49$, $R^2 = 0.014$), nor in stands of repeated disturbances ($F = 0.0004$, $P = 0.98$, $R^2 = 0.0001$ and $F = 0.05$, $P = 0.83$, $R^2 = 0.001$). The smallest FWD loadings were observed on burned sites, both fire-alone and with previous bark beetle.

Burned sites, both with and without prior bark beetle activity, had significantly smaller litter and duff depths compared to unburned sites ($F = 39.58$, $P < 0.0001$ and $F = 34.12$, $P < 0.0001$, respectively). This was the only significant difference in the pairwise comparisons as there were no significant differences between phases of bark beetle mortality and control sites, and litter and duff depths were similar among all burned sites.

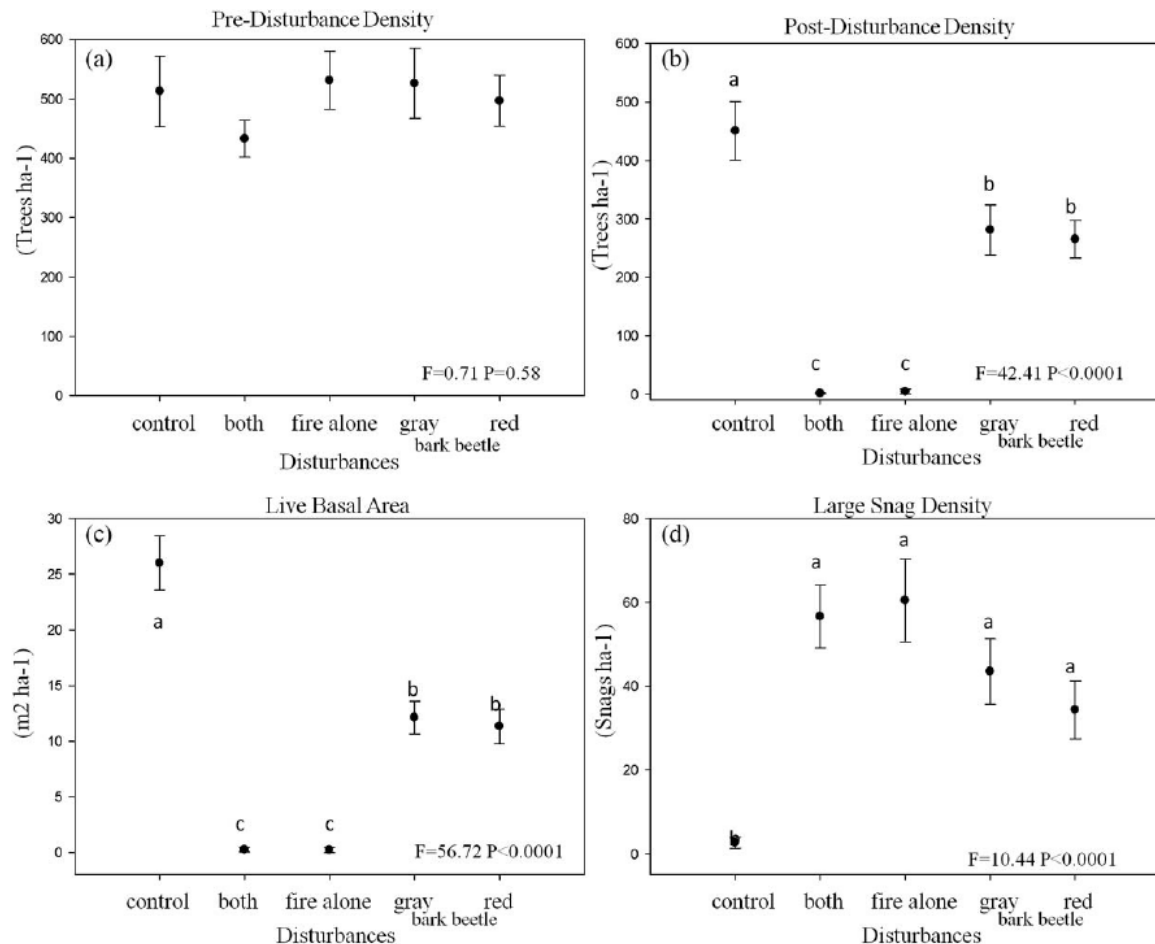


Fig. 3. Stand structure characteristics from the five disturbances across all four fires. (a) Pre-disturbance tree density (trees > 5 cm DBH), (b) post-disturbance live tree density (trees > 5 cm DBH), (c) live basal area, and (d) large snag (> 40 cm DBH) density. Error bars indicate the standard error for each disturbance type. F and P values from the ANOVA (N = 180) are also provided in each figure and letters above each point indicate results from a Tukey's HSD test.

Significant differences were observed in the multivariate analysis of combined fuels and stand structure characteristics among disturbances ($P < 0.0001$). Based on the pairwise comparisons, red and gray phase sites were not significantly different from one another ($P = 0.31$) but did vary significantly from burned sites (both fire alone and bark beetle and fire) and control sites ($P < 0.0001$). Additionally, while bark beetle followed by wildfire sites and wildfire only sites did not significantly differ from one another ($P = 0.61$), they were significantly different from all unburned sites (control,

red and gray phase sites; $P < 0.0001$). Further, control sites were significantly different from sites of any disturbance types ($P < 0.0001$).

DISCUSSION

Our empirical data suggests that in areas with high tree mortality, regardless of the agent of mortality, there are no interactive effects on post-disturbance seedling density or fuel complexes in dry mixed conifer forests of the Northern Rocky Mountains. Empirical studies on interacting disturbance types, specifically bark beetle mor-

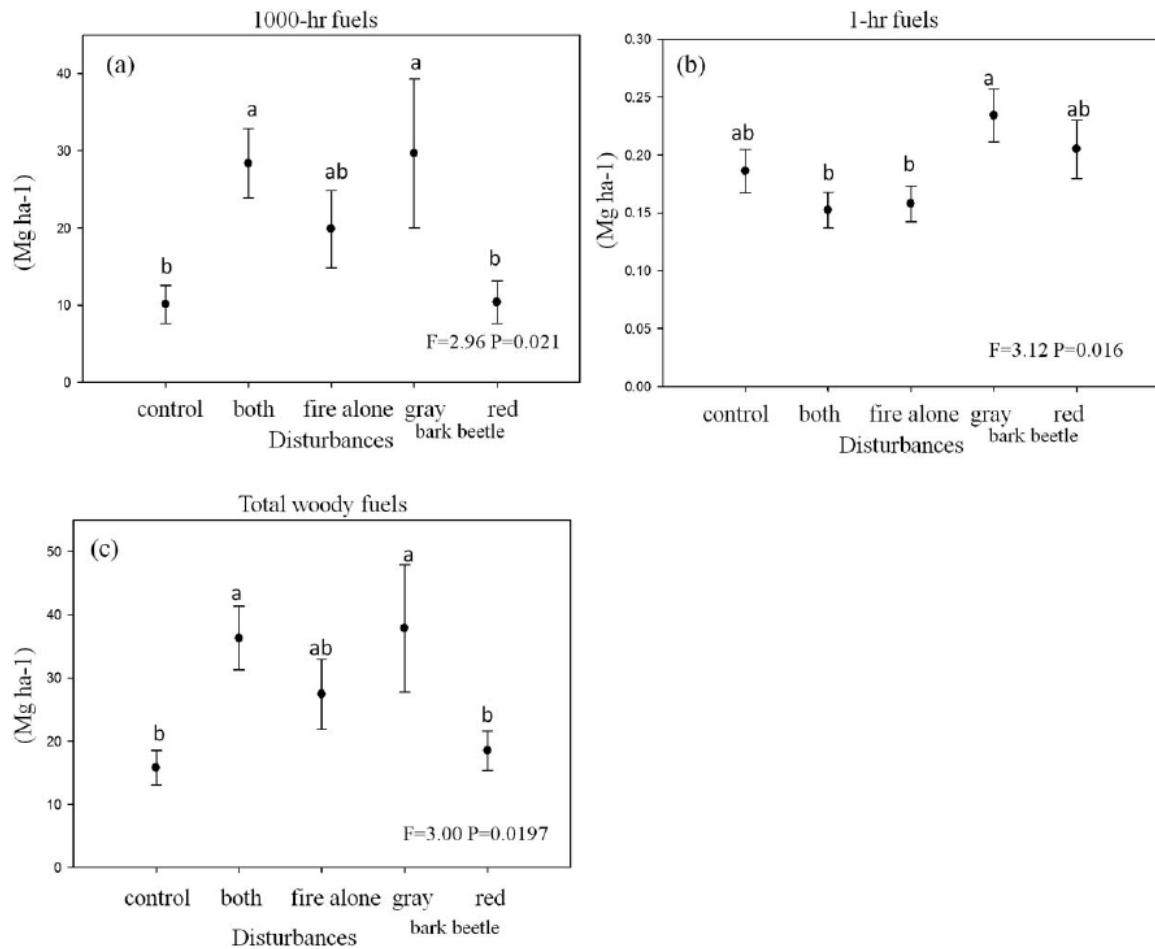


Fig. 4. Woody surface fuel loadings of those size classes with significant differences between treatment groups: (a) 1000-hr fuel loadings (>7.62 cm diameter), (b) 1-hr fuel loadings (<0.64 cm diameter) and (c) total woody fuels. Error bars indicate the standard error for each disturbance type. F and P values from the ANOVA (N = 180) are also provided in each figure and letters above each point indicate results from a Tukey's HSD test.

tality followed by wildfire events, in dry-mixed conifer forests have been largely limited to simulation modeling. We show that post-fire tree seedling age distribution was impacted by interacting disturbances. Still, total seedling density did not vary as a result of repeated disturbances, which did not support our hypothesis that seedling density would be reduced following repeat disturbances. Conversely, as predicted, there was no relationship between bark beetle outbreaks before wildfires and subsequent fuel loading post-wildfire, suggesting that disturbances do not interact to alter fuel complexes.

Tree regeneration: mixed interactive effects of fire and bark beetles

Tree seedling density was lower in both fire alone stands and in bark beetle and wildfire stands compared to unburned stands suggesting that there is not an interactive effect of fire and bark beetles on tree seedling densities when burn severity is high. These results contradict those of Harvey et al. (2013) and Kulakowski et al. (2013) who found lower seedling densities on bark beetle and fire affected sites compared to fire alone. Differences in forest type and site productivity between studies may be the cause of the mixed results (Kramer and Johnson 1987, Shat-

ford et al. 2007). As no seedlings were detected on 95% of all burned sites, any additional reduction in seedling density from previous bark beetle activity may not be visible on these overall low regeneration sites. Unlike previous studies, we did not find observable differences in seedling density between fire alone and bark beetle and fire affected sites.

Although burned sites did not vary based on previous bark beetle activity, multiple differences in seedling densities were observed across disturbance types. The highest seedling densities on gray phase sites demonstrate the importance of time-since-outbreak on stand structure and in the establishment of tree seedlings (Collins et al. 2011). Seedling density on red phase sites was not significantly different from control sites, likely due to the recent nature of the disturbance (Collins et al. 2011, Schoennagel et al. 2012). Though seedling density was lower on burned sites compared to unburned sites, when seedlings were present the observed densities were not below recommended levels for stand replacement in this forest type (Shatford et al. 2007).

In addition to total tree seedling density, greater densities of individual species were observed within different disturbance types, likely a result of physiological differences in species. We found a greater abundance of Douglas-fir and grand fir seedlings on unburned sites compared to ponderosa pine. These results are in part explained by the relative abilities of Douglas-fir and grand fir to regenerate on both mineral soil and in litter and duff layers (Gray and Spies 1996, Malcolm et al. 2001), whereas shade-intolerance and a preference for mineral soil found after wildfires led to greater ponderosa pine seedling densities on burned sites (Romme 1982, DeLuca and Sala 2006). One of our four fires, the Rattlesnake Fire, had a higher proportion of lodgepole pine seedlings compared to the other wildfires. These sites, though identified as Northern Rocky Mountain Dry-Mesic Montane Mixed Conifer Forests using LANDFIRE ESP data, had lodgepole pine in the pre-disturbance stand structure comprising <25% of its structure and the other three fires had <5% lodgepole pine. Lodgepole pine seedling densities were greatest on burned sites due to their serotinous cones and role as an early colonizing tree species (Schwilk and Ackerly

2001, Schoennagel et al. 2003).

Although tree seedling density did not vary between fire-alone and repeated disturbances stands, there were differences in the age distribution of both Douglas-fir and grand fir seedlings. In both cases bark beetle and fire-affected stands had significantly older seedlings compared to fire-alone stands. Most seedling age distributions are right-skewed with a majority of seedlings only a year or two old, and we expect to see continued seedling establishment through time (Shatford et al. 2007) with perhaps a decline in older seedlings, because survivorship is low in young tree seedlings (Peet and Christensen 1987, Gray and Spies 1996). This pattern can be observed in our control, red, and gray phase stands. However, bark beetle and fire affected stands had a significantly older cohort of seedlings that was approximately 3–6 years old. Neither distance to seed source nor any topographic variables could explain these differences as the influence of these bottom-up factors (i.e., slope, elevation, and distance to seed source) were consistent across disturbances. A decreased seed bank from repeated disturbances may be one explanation for the reduction in continued germination of Douglas-fir seedlings (Stark et al. 2006, Harvey et al. 2013). At the time of the wildfire these repeated disturbance stands were in the gray phase, thus we expect that these stands had similarly high seedling densities as we observed in gray phase stands. The germination of an older cohort of seedlings that established in-between the two disturbances left a limited seed supply for continued germination through time (Franklin et al. 2000, Larsen and Franklin 2005). Without a sufficient seed source and lacking a seed bank these sites may have less tree regeneration and lower seedling density in future years, especially on sites where seed trees were >150 m away (Bonnet et al. 2005, Donato et al. 2009, Harvey et al. 2013).

Vegetation responses can be influenced by many factors, including distance to seed source, elevation, climate and weather. We found distance to seed source was negatively correlated with seedling density and positively correlated with elevation, similar to Donato et al. (2009) and Harvey et al. (2013). Multiple authors demonstrate a variety of additional explanations for temporal patterns of seedling establishment

including climate following disturbances (e.g., Brown and Wu 2005) and micro-climate variability (e.g., Gray and Spies 1997). However, we did not see similar age structures across all disturbance types, which would indicate favorable growing years compared to less favorable growing years if larger climatic drivers were responsible for variations in seedling establishment (Kitzberger et al. 2000). Further, we saw the same pattern within each disturbance type across a large geographic area (approximately 40,000 km²), which would not occur if more localized weather was responsible for differences in vegetation responses. Lower seedling densities in burned areas may be explained by high levels of solar radiation due to the lack of canopy cover and high soil burn severity (White et al. 1996, Zald et al. 2008). However, more detailed information in specific micro-climate drivers of seedling establishment was not measured for our sites. Disturbance types and topographic variations were likely the strongest drivers of seedling densities and age structures in this study.

***Fuel complex: higher fuels loads
with longer time-since-disturbance***

Large snag dynamics across stands illustrated the mechanism of mortality of these two disturbances and live stand structure was predicted given our a priori requirements for each disturbance. Large snag densities in burned stands did not exceed densities of stands in either phase of bark beetle-induced tree mortality even though tree mortality in burned stands was close to 100% and as low as 25% on bark beetle affected stands. The two *Dendroctonus* bark beetles species in this study specifically targeted larger diameter trees (McCullough et al. 1998, Bjork and Lindgren 2009). This resulted in a greater number of large snags to total tree mortality seen in bark beetle affected stands compared to burned stands. While the death of large, old live trees is concerning to managers, large snags also serve a key ecological role and thus the creation of large snags is not necessarily a detriment to forested ecosystems (Bull et al. 1997, Hutto 2006). Our disturbed stands averaged 35–60 snags/ha, which well exceeded the recommended snag densities for restoration of 5–8 snags/ha (Nez Perce National Forest Plan 1987, Bitterroot National Forest Plan 1989). Still, small areas of

high snag densities across the landscape may be beneficial especially during early succession following a disturbance (Haggard and Gains 2001, Kotliar et al. 2002, Hutto 2006). One large concern for areas of repeated disturbances is the potential longevity of large snags. In stands that experienced both disturbances large snags may have shorter retention times due to structural weakening from a fire after bark beetle induced mortality (Bull 1983, Bagne et al. 2008). The ecological function of snags may be short lived in areas of repeated disturbance.

High CWD loadings on gray phase sites and on burned sites (both repeated disturbances and fire alone) were consistent with a pattern of woody fuel accumulation in Douglas-fir and ponderosa pine forest types following bark beetle outbreaks (Jenkins et al. 2008, Hoffman et al. 2012) and wildfires (Passovoy and Fulé 2006, Roccaforte et al. 2012). Low CWD loadings in red phase stands were also consistent with woody fuel accumulation trajectories (Donato et al. 2013). Surface fuel accumulation is highly dependent on time-since-outbreak and minimal changes in dead down and woody surface fuel loadings occurred 1–2 years post-outbreak regardless of percent tree mortality (Donato et al. 2013).

We found a lack of relationship between bark beetle induced tree mortality and fuel loading in areas of repeated disturbances was a result of two factors. First, these repeated disturbance stands (bark beetle and fire) were in the gray phase when the wildfire occurred thus surface fuels were likely greater at the time of the wildfire (Jenkins et al. 2008, Donato et al. 2013) and a proportion of these original fuels were consumed during the wildfire. Second, and more importantly, these stands also experienced high tree mortality from a wildfire signifying the relationship of fuel loading to percent tree mortality by only one agent (bark beetles) does not adequately capture the relationship between total tree mortality and fuel loading.

Dead and down woody fuel loadings were greater on older disturbances compared to control sites, however CWD loadings did not exceed recommended loadings six years post wildfire and three to eight years post-outbreak (Brown et al. 2003). This recommended CWD loading outlined by Brown et al. (2003) is a

combination of recommended loadings for ecological benefit such as soil productivity and small mammal habitat, while not creating excessive fire hazard and soil heating if a wildfire was to occur. Thus, at this period post-disturbance, woody biomass is providing a nutrient source to these recovering forests (Bull 1994, Kennedy and Quinn 2001) without presenting an excessive fire hazard (Brown et al. 2003). At the same time, tree mortality rates were high on these sites and only 16.4% of stems had fallen at the time of the study on burned sites and 11.5% on gray sites. We expect down woody fuel accumulation to continue as currently standing snags fall in the future. CWD loadings peak approximately 8–12 years post-wildfire (Everett et al. 1999, Passovoy and Fulé 2006, Roccaforte et al. 2012) suggesting that CWD loadings will continue to rise, perhaps beyond recommended ranges, over the next several years.

Smaller diameter woody fuel loadings varied by disturbance and lower loadings were observed on burned sites compared to bark beetles affected sites, consistent with the manner of these disturbances. While wildfires consume small woody material, bark beetle-induced mortality redistributes woody material, eventually redistributing fuels from the canopy to the forest floor (Schoennagel et al. 2012, Donato et al. 2013).

Though small woody fuel loading was consistent with fuel accumulations seen in other bark beetle affected sites, our findings were variable compared to previous wildfire studies. 10- and 100-hr fuels did not increase shortly after bark beetle disturbances (when in the red phase) and 1-hr fuel loadings in both phases (red and gray) were similar to control stands, like Hoffman et al. (2012). While mean 1-hr fuel loadings were lower on burned sites compared to gray phase sites, they did not vary between burned and control sites, similar to the findings of Stevens-Rumann et al. (2012). Inconsistent with other fuels studies following wildfires (Passovoy and Fule 2006, Stevens-Rumann et al. 2012) the 10- and 100-hr fuel loads were similar on burned areas and control sites. These varying results compared to published literature may be a result of time-since disturbance due to the high proportion of standing dead material that has not yet contributed to smaller diameter downed woody fuel loading in our stands. Conversely these varied

results may be due to the Brown (1974) methods employed for these smaller woody debris size classes. These methods have come under recent scrutiny for their weakness in capturing fuel variability (Keane et al. 2012).

CONCLUSIONS

Disturbance types impacted vegetation response and fuel structure. We did not find evidence of compounding effects of bark beetles followed by wildfires on fuel accumulation nor on tree seedling densities. Rather we note that the post-disturbance landscape is largely controlled by the changes resulting from high-severity wildfires. We only looked at areas with high tree mortality (>80%) within each wildfire, thus tree mortality and fire effects following the wildfire may have masked the structural changes from previous bark beetle outbreaks. In our study the most recent disturbance, whether bark beetle mortality or wildfire, creates similar vegetation responses and downed woody surface fuel complexes regardless of the previous disturbances.

Given the lack of empirical data on these interacting disturbances, especially in dry-mixed conifer forests, multiple future studies are warranted. First, additional data collection on herbaceous and shrubby fuel components would be beneficial in improving our understanding of the fuel complex in these various disturbance types. Second, examining interactions between these two disturbances at various burn severity levels, rather than only at high severity, would address the possibility of interaction effects across the range of conditions. The impacts of bark beetle activity may be masked and could be observed in moderate to low severity burned areas by only observing high-severity areas of a wildfire. Third, given our understanding of the impact of wildfire severity at different phases of bark beetle outbreaks (Simard et al. 2012, Hoffman et al. 2013), examining the post-fire effects of bark beetle and fire interactions at different outbreak phases is critical to understanding long-term ecological trajectories. Finally, as with all post-disturbance empirical studies, observations post-disturbance limits our understanding of the initial cause of these disturbances and we cannot address questions of why these areas experi-

enced wildfire and/or bark beetle outbreaks and the reason for varying severities. We are limited to observing current stand structure and surface fuels but cannot directly address causality of these changes nor make inferences on fire behavior or outbreak dynamics. Repeated measurements studies examining changes in fire behavior and whether these changes are a result of outbreak induced changes in stand structure or fire weather would address the causal questions (Hoffman et al. 2013).

As temperatures increase with climate change both bark beetle outbreaks and fire events are expected to occur more frequently and over larger areas especially in dry mixed conifer forests (Bentz et al. 2010, Westerling et al. 2011). We expect to see increasing overlap of these disturbances in space and in quick succession. Our study increases the understanding of interactions between these disturbances in dry mixed conifer forests. This knowledge informs managers on stand specific interactions and helps inform and validate model simulations the scientific community has heavily relied on to answer questions about disturbance interactions.

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